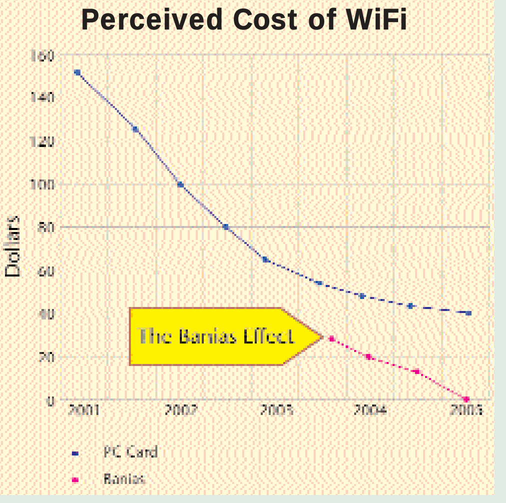
What Happens to WLAN?

Centrino has a massive significance for wireless LAN (WLAN) because of its support for the 802.11a and 802.11b wireless LAN standards, requiring only the addition of an external transceiver and modem interface. The effect of this is that a laptop manufacturer who uses the Banias chipset will have a WiFi-ready motherboard. Intel has promised to provide the transceiver in the form of a mini-PCI interface using its own silicon, codenamed Calexico.

Nick Hunn, Managing Director of TDK Systems Europe that deals with mobile connectivity technology asserts that, “Although Intel is pushing the 802.11a+b combination, I do not believe that many manufacturers will jump at the chance of incorporating it. The limited availability of 802.11a products has delayed a number of fundamental questions being asked about its global acceptance. There is still a lack of worldwide approvals for 802.11a, with very limited acceptance within Europe. In addition, whereas 802.11b is becoming deemed safe by airlines, the operating frequencies of 802.11a may well cause them concern, as they lie much closer to the frequencies where several radar systems operate.”



Banias, according to Hunn, presents a problem for AMD. The Am1772 WiFi chipset, which provides a soft baseband with a PCI interface, is less integrated than the Banias approach and will be more expensive than the Intel solution. This is good news for the laptop purchaser, who will see the cost of adding WiFi to a laptop falling significantly during 2003.

Hunn also opines that several silicon suppliers who make wireless LAN components will perish, following the decision of both Intel and AMD to provide the bulk of the circuitry on the motherboard. He points out that Centrino provides almost everything a PC maker needs to provide wireless support on the motherboard. It is not yet clear, though, what percentage of laptops will have the full RF solution placed on the motherboard. For those that do not, some manufacturers may provide connectors for a wireless LAN transceiver module to be added. Intel has already identified this, and suggested that it will supply mini-PCI transceivers for multimode in June 2003.

light notebook, making it a whole lot easier to work, play and connect on the go.

Chipsets that save power

Two chipsets (Northbridges) are being launched with the Pentium-M processor, as a part of the Centrino mobile architecture—the 855PM (Odem) and 855GM (Montana-GM). Both the 855 chipsets are essentially highly power-optimised versions of the 845 chipset. They both feature a 64-bit DDR266 memory interface and AGP 4X support, but they consume less than half the power of the 845 chipset.

The GM includes a power-saving version of Intel’s 845G graphics core, while the 855PM has no integrated graphics. Both the 855PM and GM use ICH4-M (Southbridge). Typical of any Intel Southbridge, it is responsible for managing the various power-saving mechanisms, such as Quickstart, Deep Sleep, Deeper Sleep and the enhanced SpeedStep of the Pentium-M. The processor now recognizes further modes that lie somewhere between Performance mode and Battery Optimized mode-similar to AMD’s PowerNow and Transmeta’s LongRun.

Both chipsets include a 400 MHz processor system bus and support for up to 2 GB of DDR 266 memory, along with support for USB 2.0 and Intel’s I/O Hub Architecture.

The wireless capability

The third piece of the Centrino technology is Intel’s PRO/Wireless 2100 card (formerly code-named Colexico). Notebook manufacturers must use this card (along with the 855 chipset and a Pentium-M CPU) to be able to call their notebook a Centrino system. It has been designed and validated to connect easily to 802.11b WiFi certified access points.

With Centrino mobile technology, you can connect to the Internet or a corporate network without wires or an add-on adapter card, with the integrated 802.11b wireless-LAN (WLAN) capability. WLAN uses radio waves to wirelessly connect computers to each other, to the Internet or to wired networks.

Intel Centrino mobile technology supports 802.11b WLAN standards and enables wireless connectivity from WLAN networks-including hotspots worldwide. An increasingly popular way to work and play on the go, hotspots provide WLAN service (for free or for a fee) from a wide variety of public meeting areas, including coffee shops, airport lounges and convention centres. To use these hotspots, your computer must be configured with WiFi certified technology so that you can connect with other WiFi certified products. With Intel Centrino mobile technology, you’ve got the WiFi certification you need.

Incidentally, WiFi maxes out at 11 Mbps over the 2.4 GHz frequency. The